

Glossary: 02VariationalCalculus

A bilingual glossary of key concepts introduced in Variational Calculus (Chapter 6 of Taylor's *Classical Mechanics*).

English	中文	Brief explanation
Single-variable function	单变量函数	A function $f : \mathbb{R} \rightarrow \mathbb{R}$ studied in ordinary differential calculus.
Stationary point	驻点	A point where $df/dx = 0$; the graph is locally flat to first order.
Locally flat	局部平坦	At a stationary point, an infinitesimal change in the variable produces no first-order change in the function.
Calculus of variations	变分法	The branch of mathematics that looks for a function making a functional stationary.
Functional	泛函	A rule that takes an entire function as input and returns a single number, e.g., $F[y]$.
Square-bracket notation	方括号记号	Writing $F[y]$ to emphasize that the input is a function, not a number.
Input function	输入函数	The function $y(x)$ that is varied in a functional.
Output number	输出数值	The single number returned by evaluating a functional on a chosen function.
Smooth function	光滑函数	A function with enough continuous derivatives for the calculation at hand.
Action	作用量	The functional $S[q] = \int_{t_1}^{t_2} L(t, q, \dot{q}) dt$ in Lagrangian mechanics.
Lagrangian	拉格朗日量	The function $L(t, q, \dot{q})$ whose integral gives the action.
Hamilton's principle	哈密顿原理	The physical path $q(t)$ makes the action stationary, $\delta S = 0$.
Variation	变分	A small change in the input function, $y_\varepsilon(x) = y(x) + \varepsilon \eta(x)$.
Variation function	变分函数	The perturbation shape $\eta(x)$ in a variation; it gives the direction in function space.
Small parameter	小量	The number ε controlling the size of a variation.

Direction in function space	函数空间中的方向	The role played by $\eta(x)$ when moving away from $y(x)$ in the space of functions.
Fixed endpoints	固定端点	Boundary conditions requiring all allowed curves to pass through the same two points, $\eta(x_1) = \eta(x_2) = 0$.
Admissible variation	容许的变分	A variation that satisfies the required boundary and smoothness conditions.
First variation	一阶变分	The linear part of the change in a functional, $\delta F[y; \eta] = \left. \frac{d}{d\varepsilon} F[y + \varepsilon\eta] \right _{\varepsilon=0}$.
Directional derivative	方向导数	The rate of change of a functional at y in the direction η .
Variation of a function	函数的变分	$\delta y(x) = \eta(x)$, the first-order change in y .
Variation of a derivative	导数的变分	$\delta y'(x) = \eta'(x)$; equivalently $\delta y' = d(\delta y)/dx$.
Commute variation and derivative	变分与求导可交换	Because x is held fixed, $\delta(dy/dx) = d(\delta y)/dx$.
Stationary function	驻函数、驻值函数、使泛函取驻值的函数、驻值曲线	A function $y(x)$ for which $\delta F[y; \eta] = 0$ for every admissible η .
Integrand	被积函数	The function $f(x, y, y')$ that appears inside the integral defining a functional.
Partial derivative with respect to y	对 y 的偏导数	$\partial f / \partial y$, holding x and y' fixed in the integrand.
Partial derivative with respect to y'	对 y' 的偏导数	$\partial f / \partial y'$, holding x and y fixed in the integrand.
Total derivative with respect to x	对 x 的全导数	d/dx applied after substituting $y = y(x)$ and $y' = y'(x)$.
Euler-Lagrange equation	欧拉-拉格朗日方程	The necessary condition $\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = 0$ for a stationary function.
Twice-differentiable function	二阶可微函数	A function whose second derivative exists; often assumed for stationary solutions.
C^2 function	C^2 函数 / 二阶连续可微函数	A function with continuous second derivative.
Boundary term	边界项	A term generated by integration by parts that vanishes when endpoints are fixed.

Integration by parts	分部积分	Technique used to move the derivative off η' onto f in the first variation.
Fundamental lemma of the calculus of variations	变分法基本引理	If $\int_{x_1}^{x_2} \eta(x)g(x) dx = 0$ for all admissible η , then $g(x) = 0$.
Independent variable	自变量, “自”己可以独立取值的变量	The variable that is not varied, usually x or t .
Dependent variable	因变量, “因”自变量变化而变化的量	The function that is varied, usually $y(x)$ or $q(t)$.
Boundary condition	边界条件	Conditions imposed on the solution at the endpoints, e.g., $y(x_1) = y_1$.
Beltrami identity	贝尔特拉米恒等式	The first integral $f - y' \partial f / \partial y' = C$ when f has no explicit x -dependence.
First integral	第一积分	A conserved quantity obtained by integrating the Euler–Lagrange equation once.
Constant of integration	积分常数	An undetermined constant appearing when an equation is integrated.
Arc length element	弧长元	The infinitesimal length $ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + y'^2} dx$ of a curve.
Length functional	长度泛函	The functional $L[y] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx$ for a plane curve.
Plane curve	平面曲线	A curve lying in a two-dimensional plane, often written as $y = y(x)$.
Shortest path problem	最短路径问题	The variational problem of finding the curve of minimum length between two points.
Fermat's principle	费马原理	Light takes a path that makes the travel-time functional stationary.
Refractive index	折射率	The quantity $n(x, y)$ such that the speed of light in a medium is $v = c/n$.
Travel-time functional	传播时间泛函	The functional $T[y] = \frac{1}{c} \int n(x, y) \sqrt{1 + y'^2} dx$ for a light ray.
Principle of least time	最短时间原理	Fermat's statement that light follows the path of least travel time.
Brachistochrone problem	最速降线问题	Finding the curve of fastest descent for a bead sliding without friction under gravity.

Cycloid	摆线 / 旋轮线	The curve traced by a point on the rim of a rolling wheel; the brachistochrone solution.
Isochronous	等时的	Having a period independent of amplitude; oscillations on a cycloid are exactly isochronous.
Parametric form	参数形式	Representing a curve by $x = x(u)$, $y = y(u)$ using a parameter u instead of $y = y(x)$.
Parameter	参数	An auxiliary variable, such as u , used to describe a curve in parametric form.
Euler-Lagrange equation for several variables	多变元欧拉-拉格朗日方程	One equation per dependent variable: $\partial f / \partial q_i - \frac{d}{du}(\partial f / \partial \dot{q}_i) = 0$.